

GOING REACTIVE:

New and Old Ideas for Your
21st Century Architectures

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TECHNICAL DIRECTOR



JONAS **BONÉR**

CTO



AGENDA

AGENDA

1. New Challenges for a New Era

AGENDA

1. New Challenges for a New Era
2. Go Reactive

AGENDA

1. New Challenges for a New Era
2. Go Reactive
3. Event-Driven

AGENDA

1. New Challenges for a New Era
2. Go Reactive
3. Event-Driven
4. Scalable

AGENDA

1. New Challenges for a New Era
2. Go Reactive
3. Event-Driven
4. Scalable
5. Resilient

AGENDA

1. New Challenges for a New Era
2. Go Reactive
3. Event-Driven
4. Scalable
5. Resilient
6. Responsive

IT'S WEBSCALE!

TELECOMS WAS WEBSCALE BEFORE THE WEB WAS INVENTED



DEMANDS AND EXPECTATIONS FOR APPLICATIONS HAVE CHANGED
DRAMATICALLY IN RECENT YEARS

IT'S WEBSCALE!

TELECOMS WAS **REACTIVE** BEFORE THE **MANIFESTO**
WAS FORMALISED



DEMANDS AND EXPECTATIONS FOR APPLICATIONS HAVE CHANGED
DRAMATICALLY IN RECENT YEARS

HETEROGENEOUS MULTI-CORE HARDWARE IS HERE TO STAY

Different cores doing different things
CPUs, GPUs, FPGA



Parallella Board

Dual core ARM processor + FPGA

1GB RAM

16 or 64 core Epiphany co-processor

Gigabit Ethernet

THE FASTEST COMPUTER IN THE WORLD!



TIANHE-2, CHINESE NATIONAL UNIVERSITY OF DEFENCE TECHNOLOGY

33.86 petaflops/s (November, 2013) 16,000 Nodes; each with 2 Ivy Bridge multicores and 3 Xeon Phis

3,120,000 X86 CORES IN TOTAL

WEATHER FORECAST?

London, United Kingdom Weather

Right Now



9°C

Cloudy

Feels Like 6°C

❤️ Love!

Yuck!

Today

Hourly

Weekend

10-Day Forecast

Updated: last updated about 1 hour(s) ago

Tonight

29. Nov



11°C

Observed
High
12:05

6°C

Overnight
Low

CHANCE OF
RAIN
0%

WIND

NW at 15
mph

[Expand Details](#)



Tomorrow

30. Nov



8°

High

2°
Overnight
Low

CHANCE OF
RAIN
0%

WIND

NNW at 14
mph

[Expand Details](#)



CLOUD COMPUTING

ON DEMAND SERVICE
BROAD NETWORK ACCESS
RESOURCE POOLING
RAPID ELASTICITY
MAINTAINABILITY



THE INTERNET OF THINGS

GSM ASSOCIATION PREDICTS WE WILL HAVE 24 BILLION DEVICES BY 2020

ERICSSON CEO PREDICTS 50 BILLION INTERNET CONNECTED DEVICES BY 2020.

MORGAN STANLEY FEELS THAT NUMBER CAN ACTUALLY BE AS HIGH AS 75 BILLION



Source: GSMA, Ericsson, Morgan Stanley

BIG DATA

THE FASTER YOU ANALYSE DATA, THE GREATER ITS PREDICTIVE VALUE

MAINTAIN ONE COPY OF YOUR DATA.

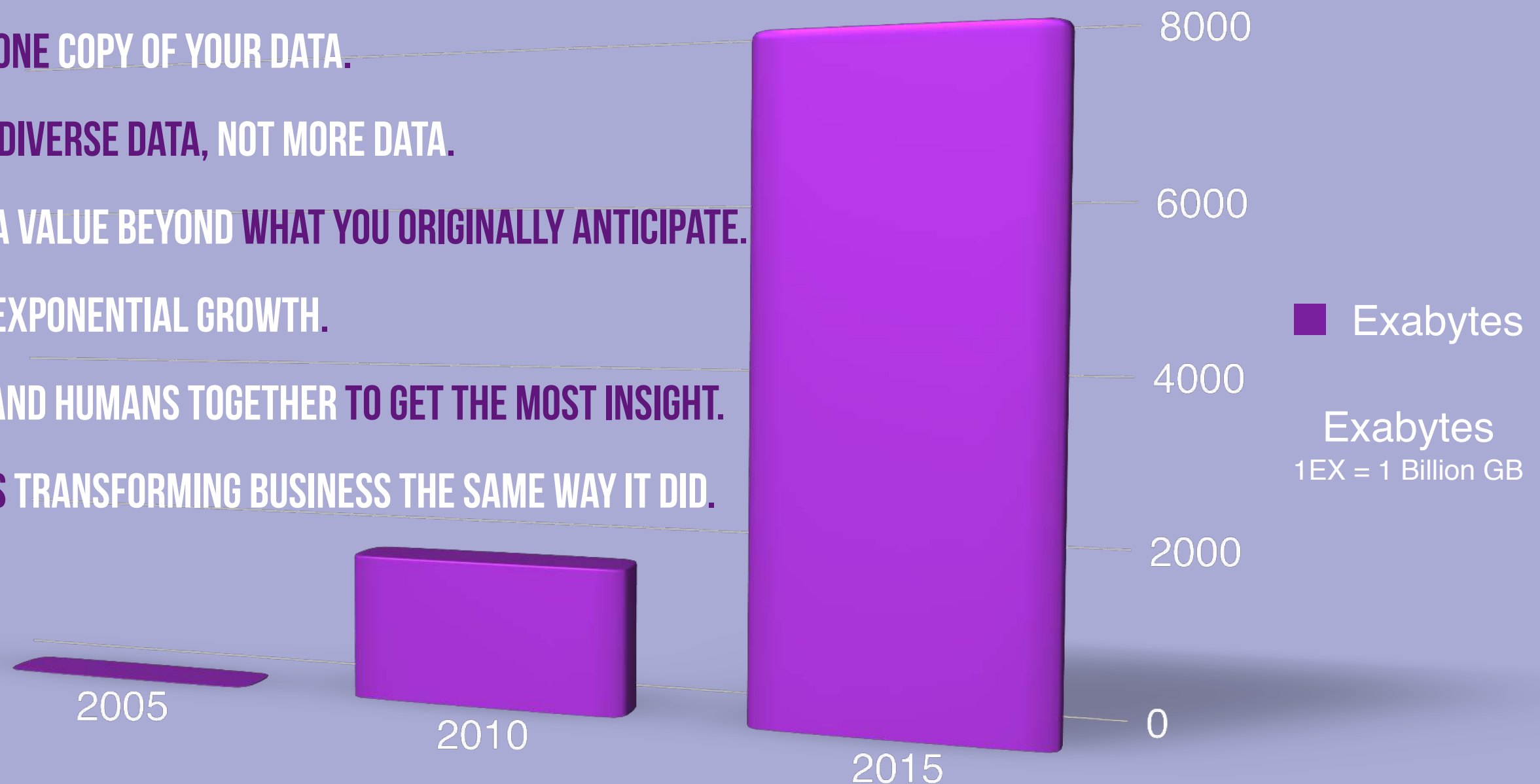
USE MORE DIVERSE DATA, NOT MORE DATA.

DATA HAS A VALUE BEYOND WHAT YOU ORIGINALLY ANTICIPATE.

PLAN FOR EXPONENTIAL GROWTH.

PUT DATA AND HUMANS TOGETHER TO GET THE MOST INSIGHT.

BIG DATA IS TRANSFORMING BUSINESS THE SAME WAY IT DID.



Source: IDC

SOFTWARE DEFINED NETWORKING

CONNECTS COMPUTERS
TOGETHER IN A WAY WE HAVE
NEVER DONE BEFORE

OPENFLOW IS THE
PREVAILING STANDARD

DEFINES THE PROTOCOL
BETWEEN THE CONTROLLER

AND THE SWITCH

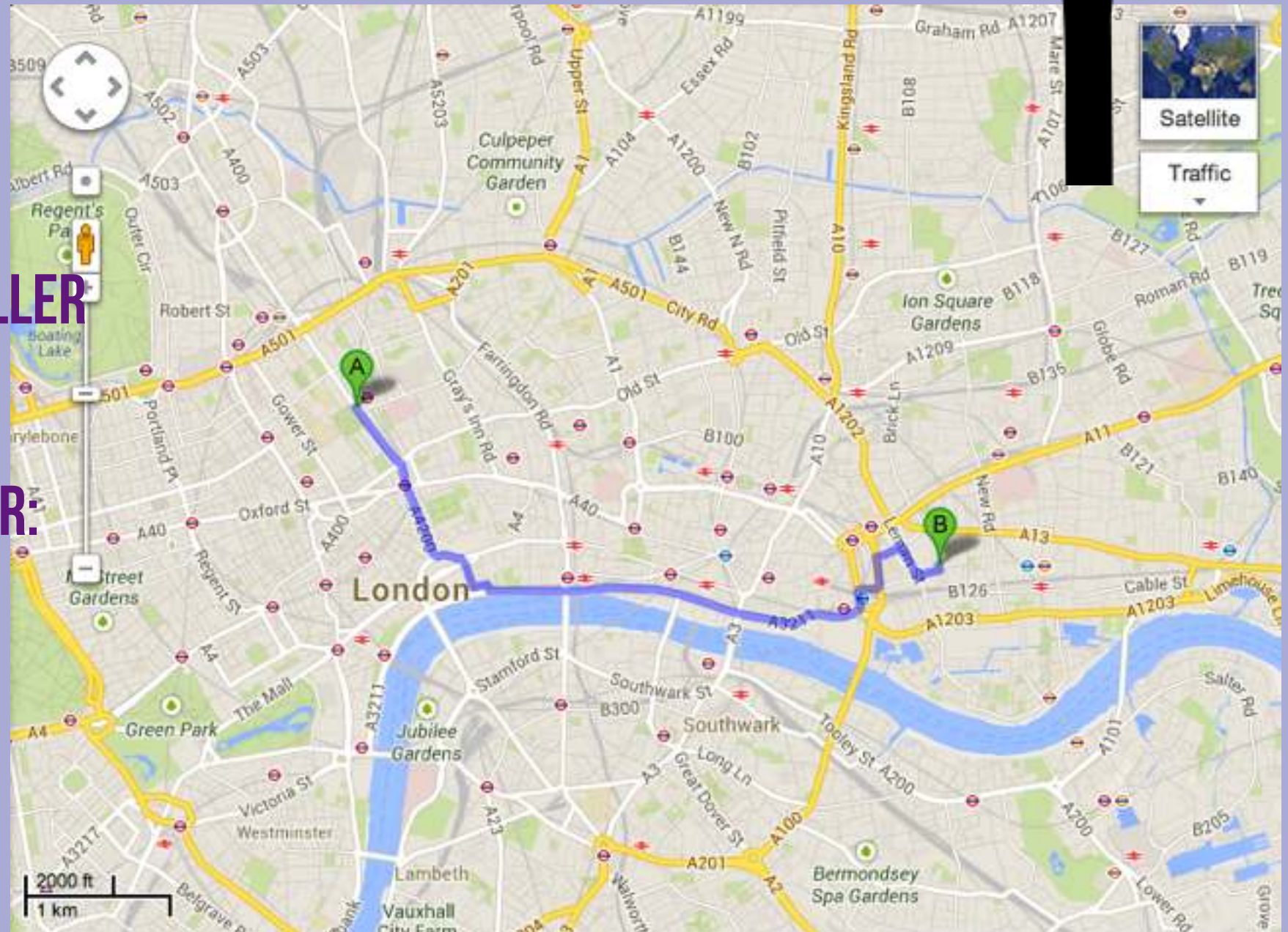
ALLOWS US TO OPTIMISE FOR:

THROUGHPUT

LATENCY

SCALABILITY

UTILISATION



SOFTWARE DEFINED NETWORKING

SOFTWARE DEFINED NETWORKING

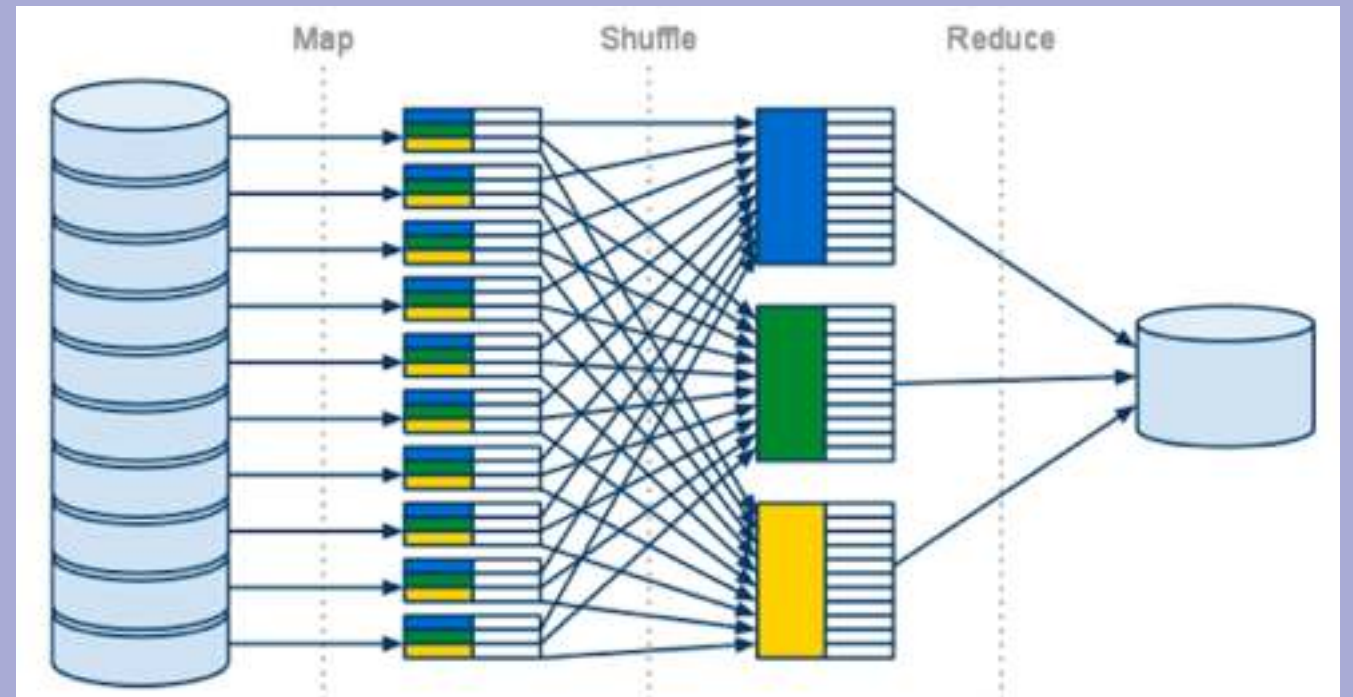
IN MAP-REDUCE **CLUSTERS**

SCALING NETWORKS **IN THE INTERNET OF THINGS**

SECURE COMMUNICATION **IN MASSIVELY DISTRIBUTED SYSTEMS**

SHIFTING AROUND BIG DATA

SOFTWARE DEFINED NETWORKING



IN MAP-REDUCE **CLUSTERS**

SCALING NETWORKS **IN THE INTERNET OF THINGS**

SECURE COMMUNICATION **IN MASSIVELY DISTRIBUTED SYSTEMS**

SHIFTING AROUND BIG DATA

REACTIVE

“READILY RESPONSIVE TO A STIMULUS”

- MERRIAM WEBSTER

THE FOUR TRAITS OF **REACTIVE**

THE FOUR TRAITS OF **REACTIVE**

EVENT-DRIVEN

THE FOUR TRAITS OF **REACTIVE**

SCALABLE

EVENT-DRIVEN

THE FOUR TRAITS OF **REACTIVE**

SCALABLE

RESILIENT

EVENT-DRIVEN

THE FOUR TRAITS OF **REACTIVE**

RESPONSIVE

SCALABLE

RESILIENT

EVENT-DRIVEN

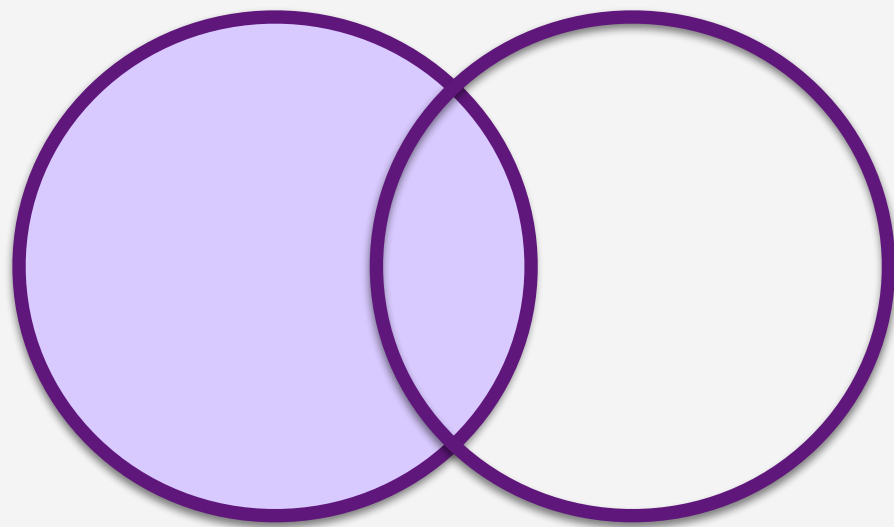
EVENT-DRIVEN

“THE FLOW OF THE PROGRAM IS DETERMINED BY EVENTS”

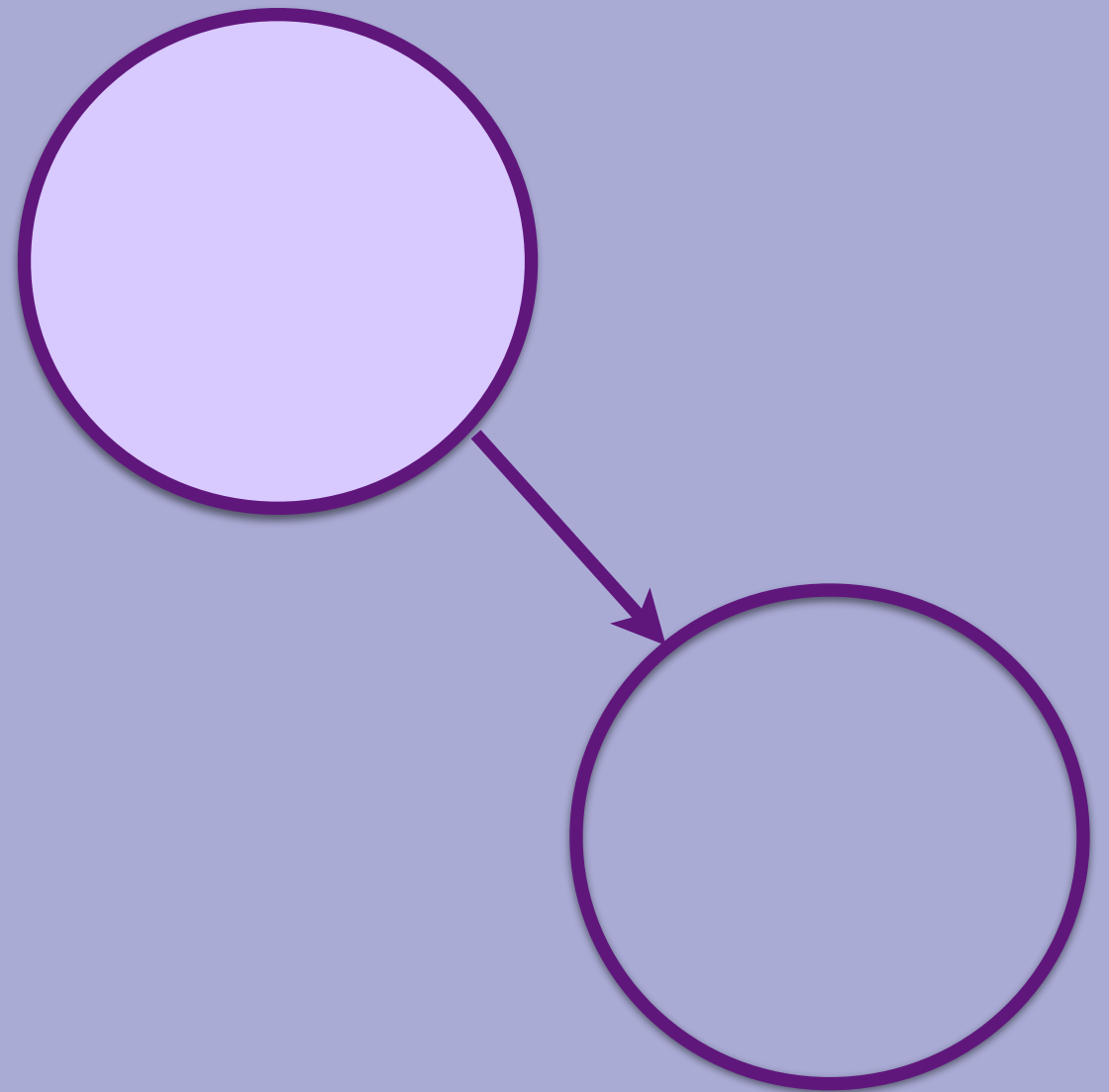
- WIKIPEDIA

TWO WAYS TO DO CONCURRENCY

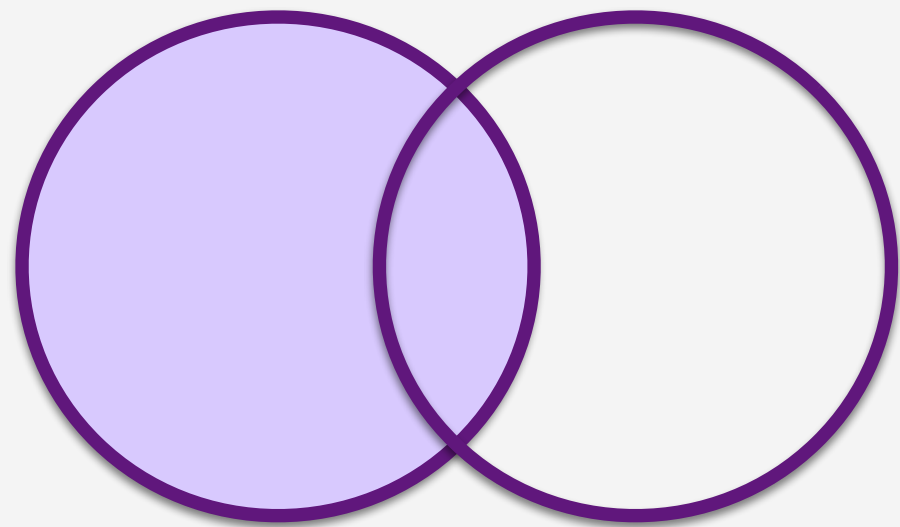
MUTABLE STATE



IMMUTABLE STATE



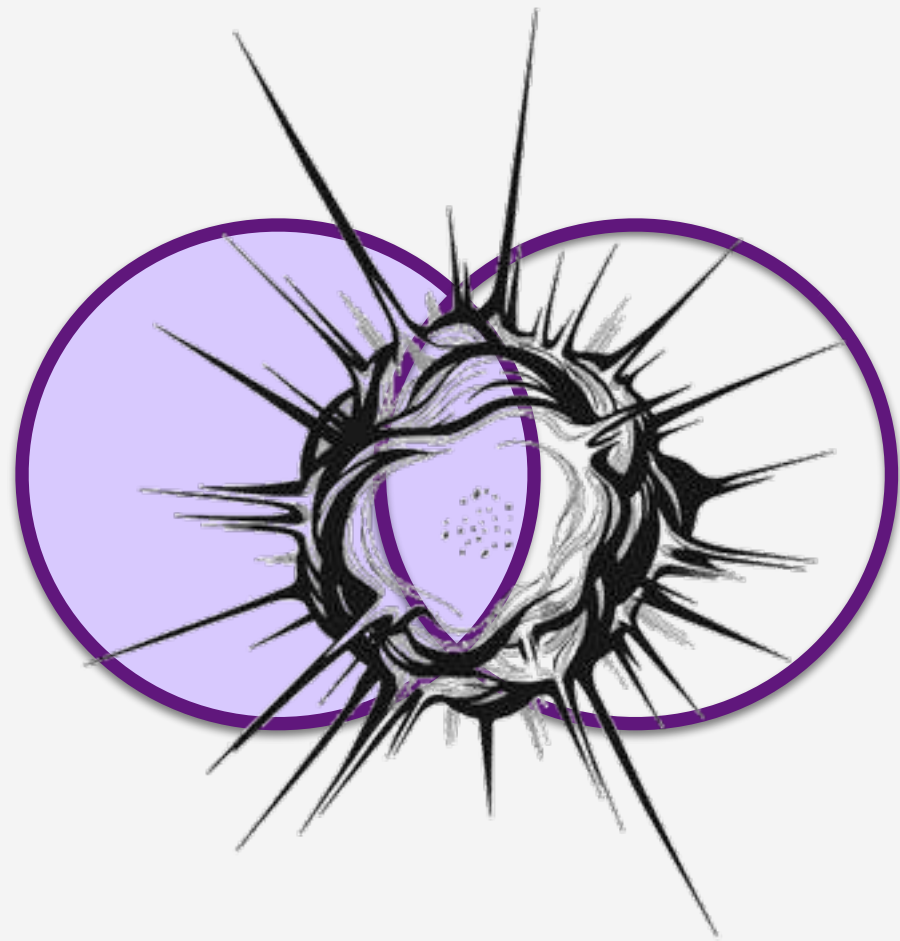
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**PROBLEM 1 WITH MUTABLE
STATE:**

**YOUR PROGRAM CRASHES
WHILST EXECUTING IN THE
CRITICAL SECTION...**

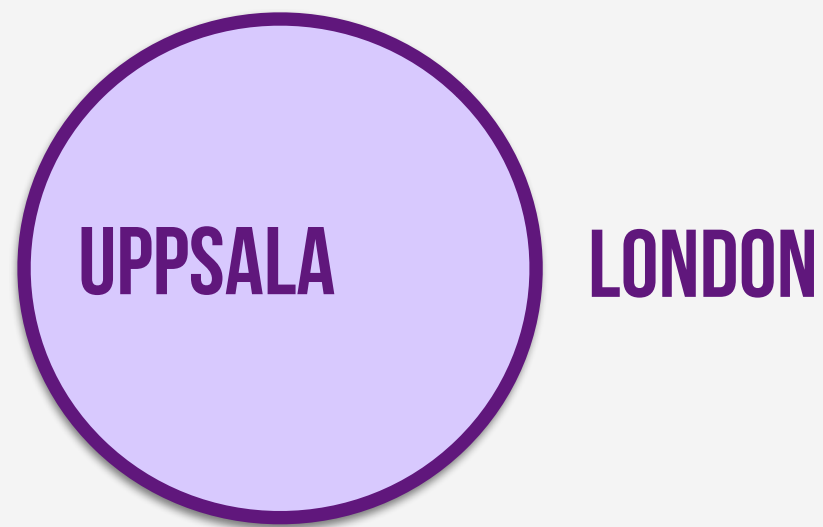
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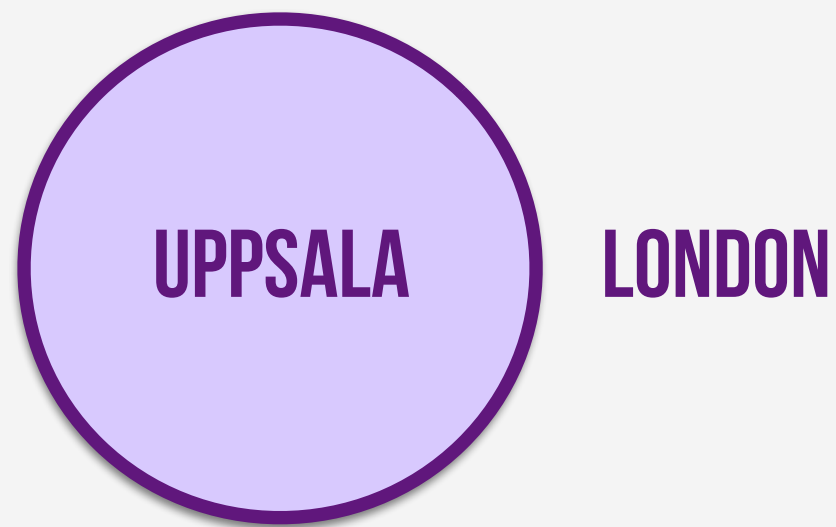
TWO WAYS TO DO CONCURRENCY



**PROBLEM 2 WITH
MUTABLE STATE:**

**WHERE DO YOU LOCATE
YOUR STATE...**

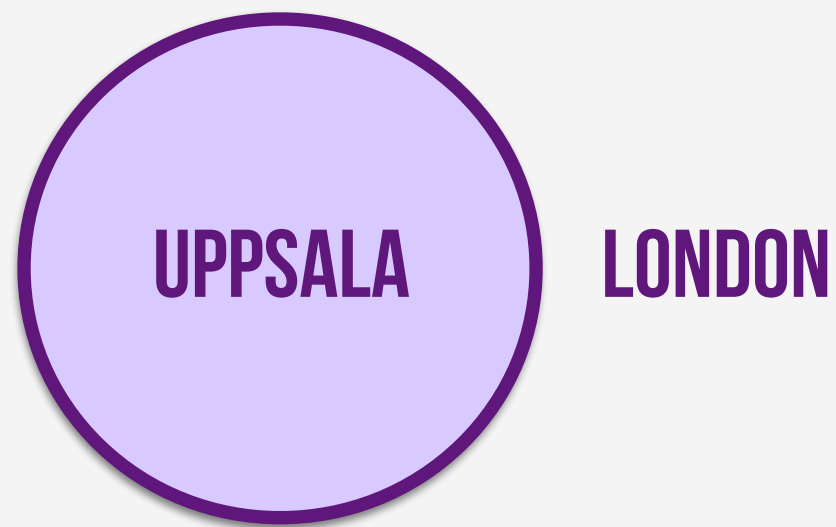
TWO WAYS TO DO CONCURRENCY



**PROBLEM 3 WITH
MUTABLE STATE:**

**WHAT HAPPENS IF YOUR
NETWORK CONNECTIVITY
FAILS...**

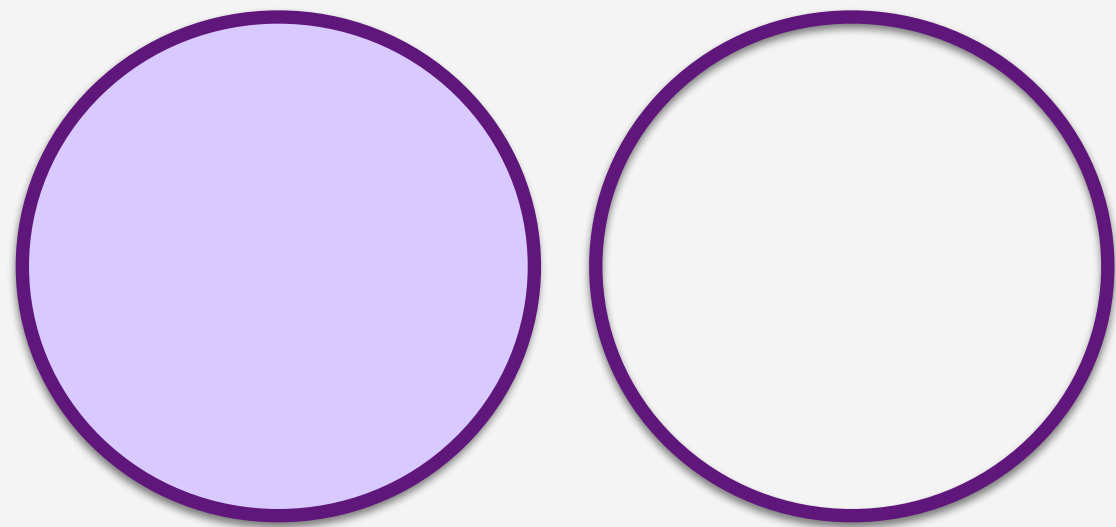
TWO WAYS TO DO CONCURRENCY



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MUTABLE STATE:**

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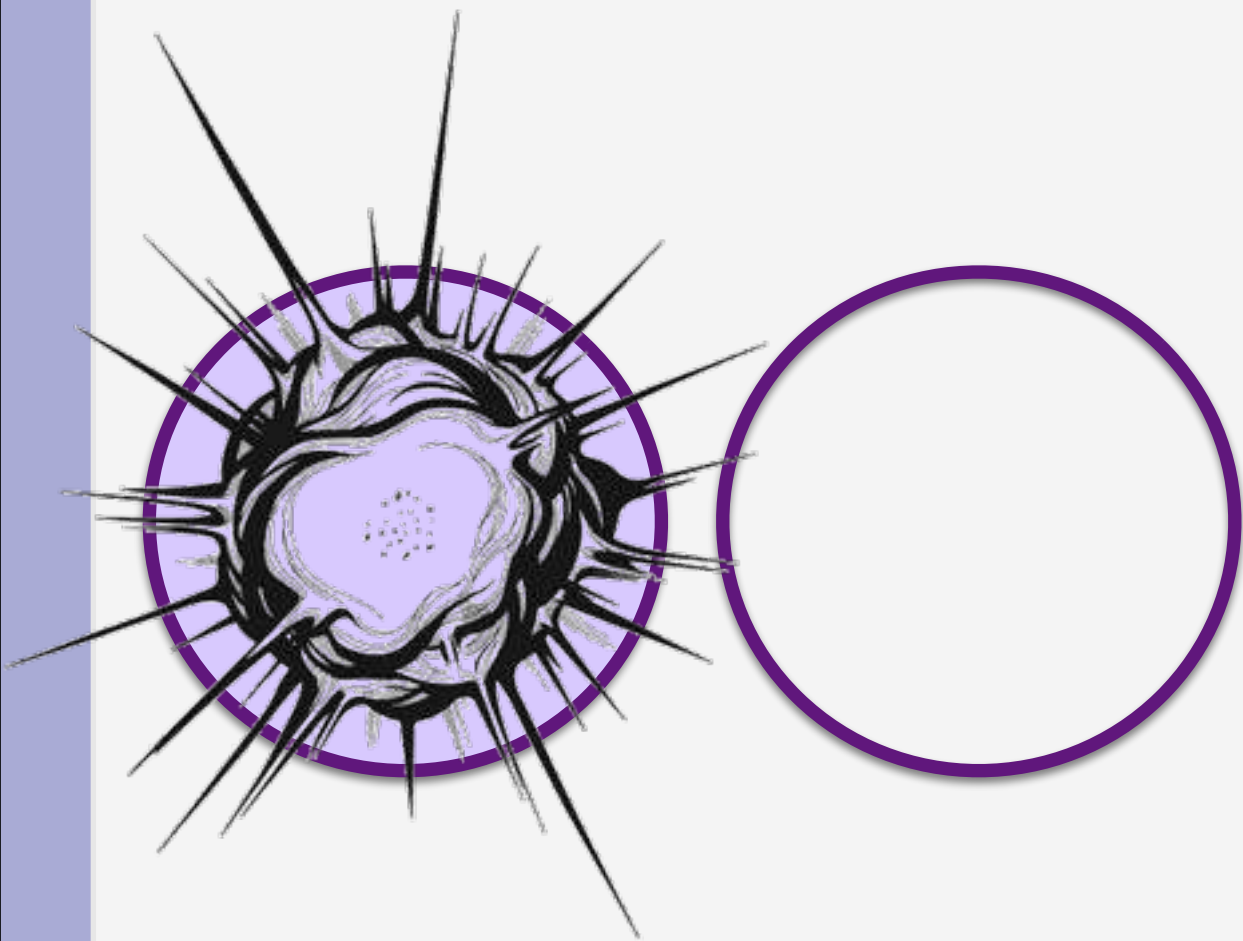
TWO WAYS TO DO CONCURRENCY



**PROBLEM 1 WITH
IMMUTABLE STATE:**

**YOUR PROGRAM CRASHES WHILST
EXECUTING IN THE CRITICAL
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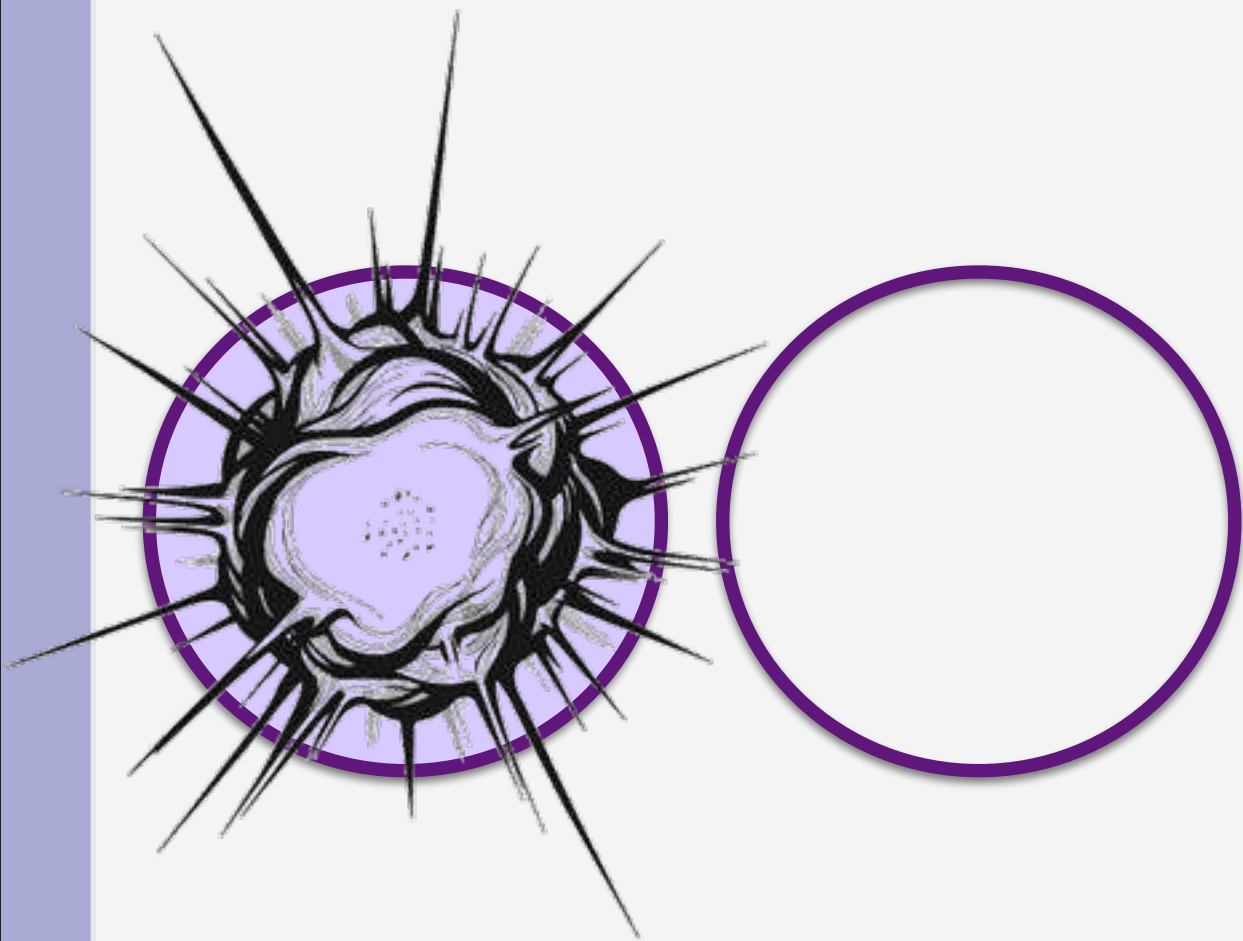
TWO WAYS TO DO CONCURRENCY



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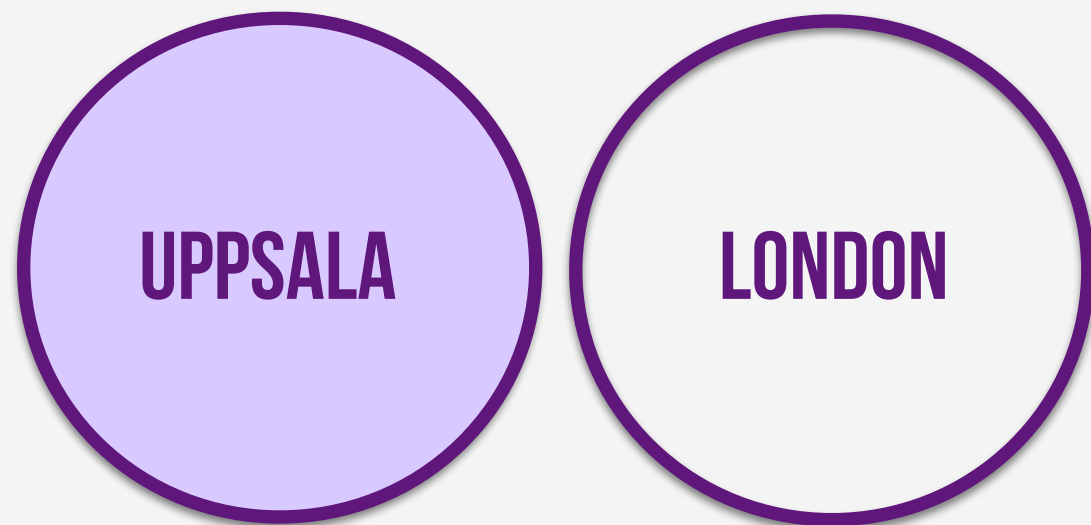


PROBLEM 1 WITH IMMUTABLE STATE:

YOUR PROGRAM CRASHES WHILST
EXECUTING IN THE CRITICAL
SECTION...

**YOUR STATE DOES NOT GET
CORRUPTED.**

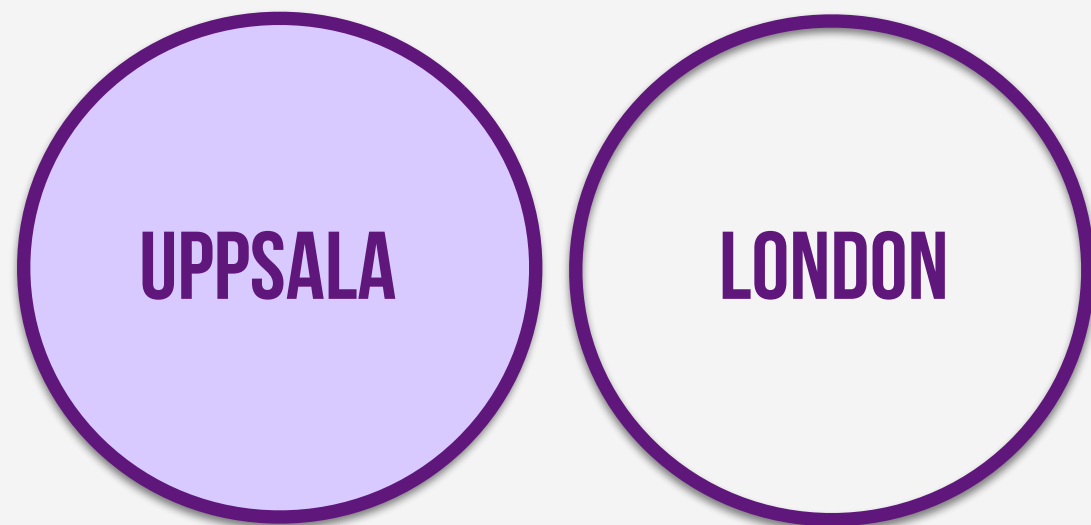
TWO WAYS TO DO CONCURRENCY



**PROBLEM 2 WITH
IMMUTABLE STATE:**

WHERE DO YOU LOCATE YOUR STATE...

TWO WAYS TO DO CONCURRENCY



**PROBLEM 2 WITH
IMMUTABLE STATE:**

WHERE DO YOU LOCATE YOUR STATE...

**YOU DO NOT LOCATE STATE,
YOU COPY IT.**

TWO WAYS TO DO CONCURRENCY



**PROBLEM 3 WITH
IMMUTABLE STATE:**

**WHAT HAPPENS IF YOUR
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FAILS...**

TWO WAYS TO DO CONCURRENCY



PROBLEM 3 WITH IMMUTABLE STATE:

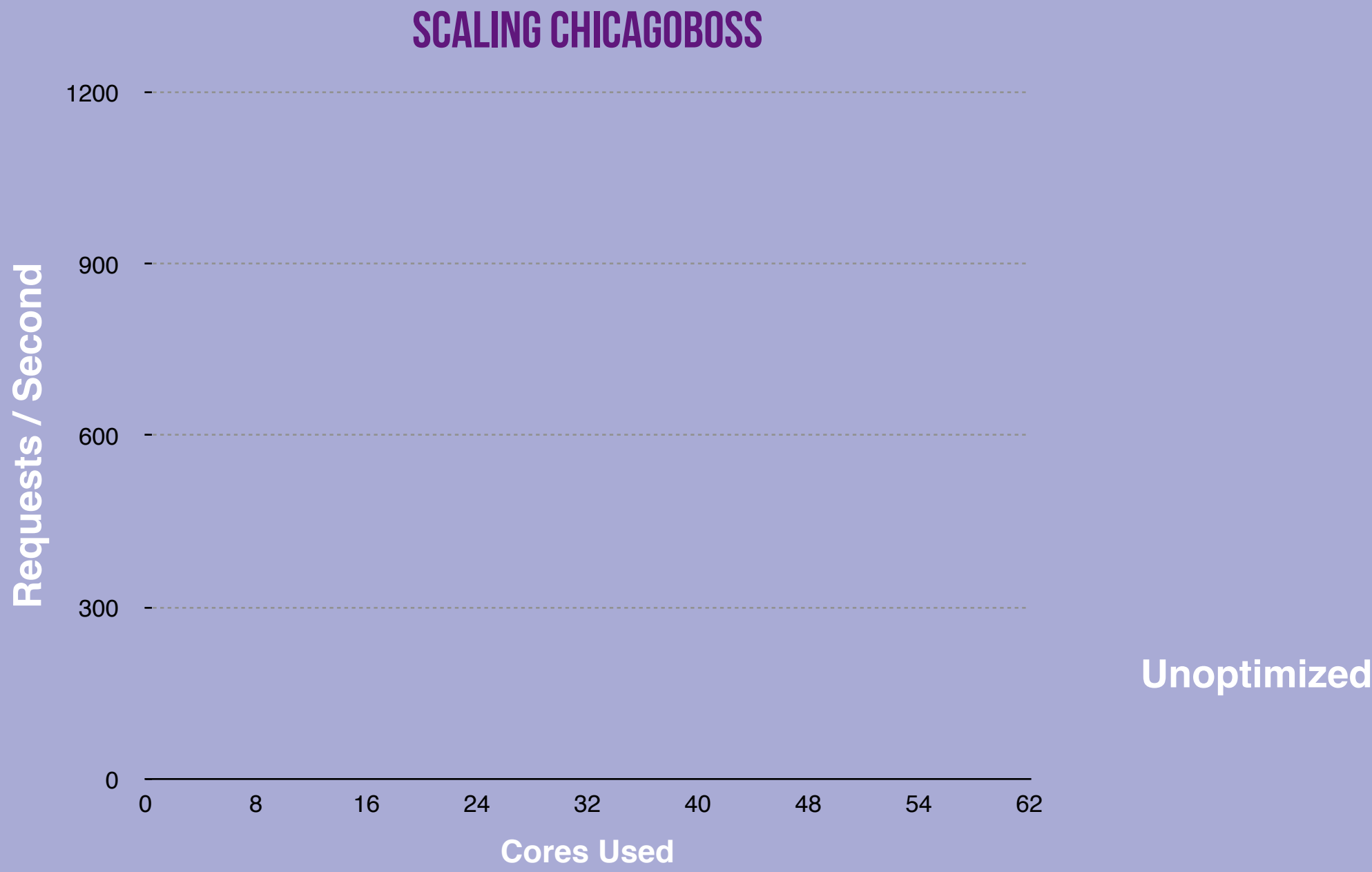
WHAT HAPPENS IF YOUR
NETWORK CONNECTIVITY
FAILS...

**MAKE SURE YOUR ALGORITHMS
AND DATABASES
HANDLE NETWORK SPLITS!**

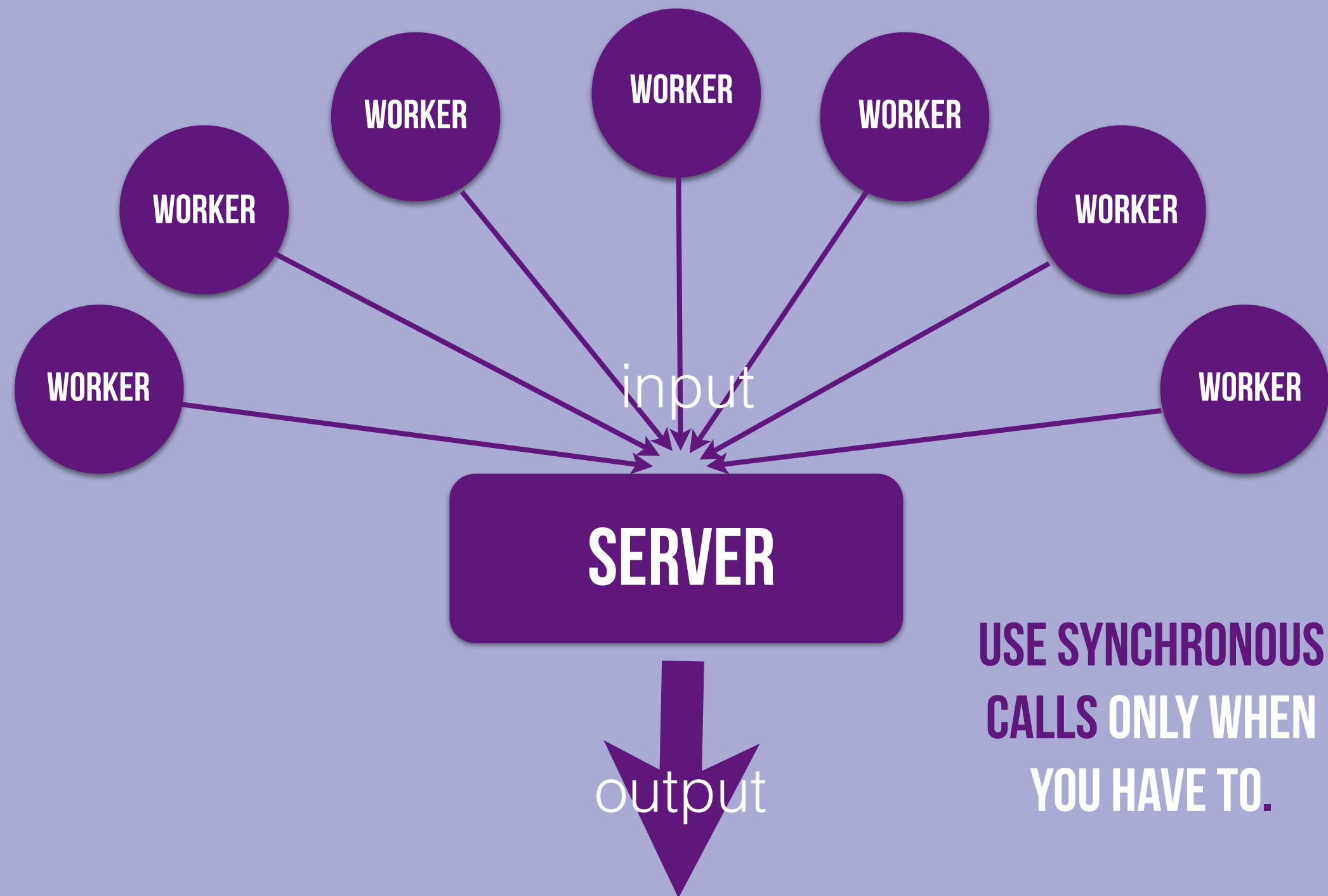
BEWARE OF CONTENTION

- IN YOUR APPLICATION, ON MULTI-CORE ARCHITECTURES AND DISTRIBUTED SYSTEMS
- YOUR SYSTEM WILL BE AS FAST AS ITS SLOWEST COMPONENT
- NEVER BLOCK
- USE ASYNCHRONOUS MESSAGE PASSING
- USE LOCK-FREE CONCURRENCY

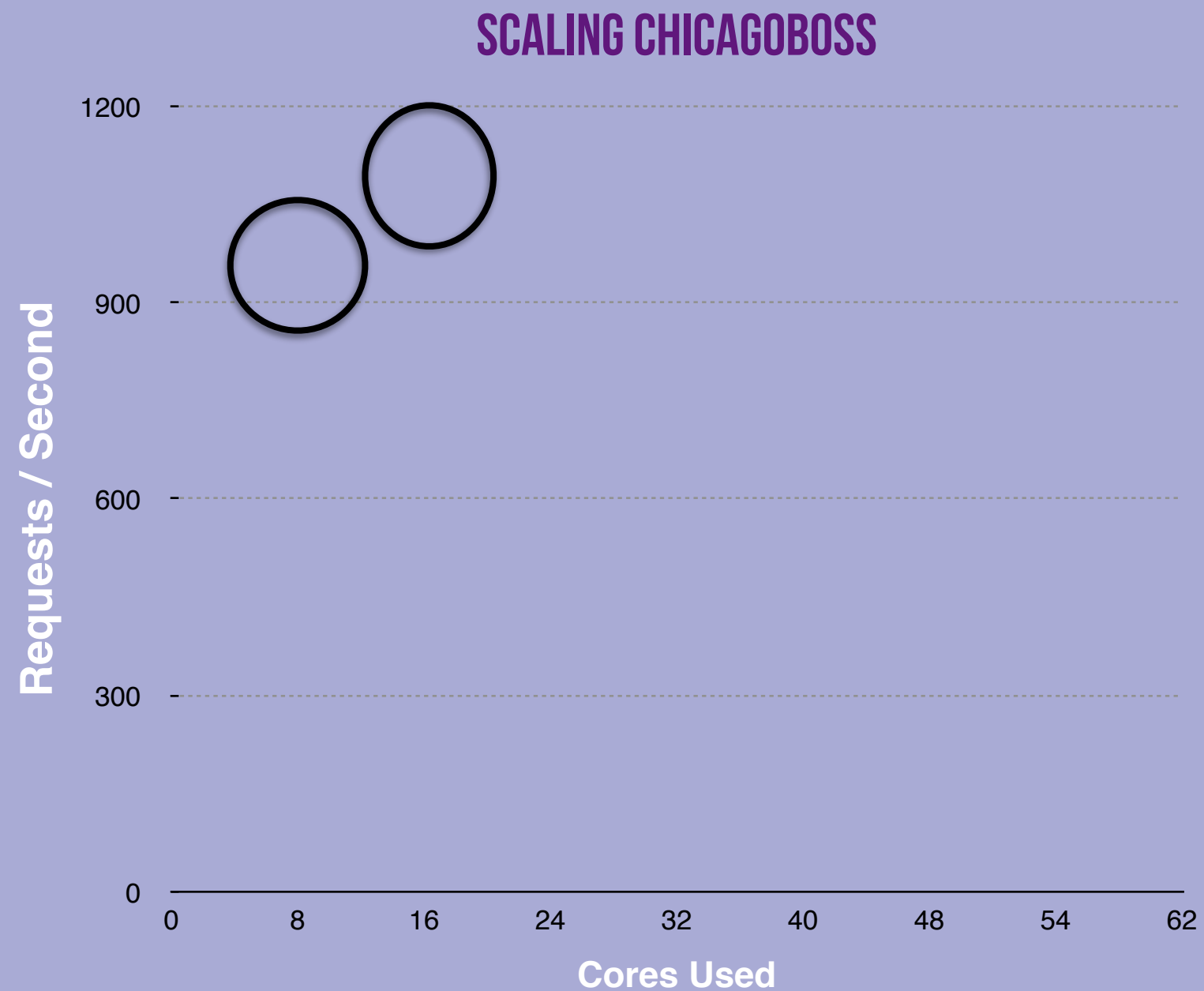
NOW FOR THE BOTTLENECKS



NOW FOR THE BOTTLENECKS...



NOW FOR THE BOTTLENECKS...



Unoptimized
Optimized

ACTORS

- Isolated lightweight event-based processes
- Each actor has a mailbox (message queue)
- Communicates through asynchronous & non-blocking message passing
- Location transparent (distributable)
- Supervision-based failure management

FUTURES/DATAFLOW

- Allows you to spawn concurrent computations and work with the not yet computed results
- Write-once, Read-many
- Freely sharable
- Allows non-blocking composition
- Build in model for managing failure

REACTIVE EXTENSIONS (RX)

- **Extend Futures with the concept of a Stream**
- **Composable in a type-safe way**
- **Observable $\Rightarrow$ Push Collections**
 - **onNext(T), onError(E), onCompleted()**
 - **Compared to Iterable $\Rightarrow$ Pull Collections**

SCALABLE

“CAPABLE OF BEING EASILY EXPANDED OR UPGRADED ON DEMAND”

- MERRIAM WEBSTER

**DISTRIBUTED
COMPUTING
IS THE NEW
NORMAL**

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COMPUTING
IS THE NEW
NORMAL**

**YOU ALREADY HAVE A
DISTRIBUTED SYSTEM,
WHETHER
YOU WANT IT OR NOT**

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Mobile

NOSQL Databases

SQL Replication

Cloud Services

**What is the
ESSENCE OF
DISTRIBUTED
COMPUTING?**

**What is the
ESSENCE OF
DISTRIBUTED
COMPUTING?**

**It's to try to
OVERCOME**

- **Information travels at
THE SPEED OF LIGHT**
- **Independent things
FAIL INDEPENDENTLY**

WHY DO WE NEED IT?

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SCALABILITY

**When you outgrow
the resources of
a single node**

WHY DO WE NEED IT?

SCALABILITY

When you outgrow
the resources of
a single node

AVAILABILITY

Providing resilience if
one node fails

WHY DO WE NEED IT?

SCALABILITY

When you outgrow
the resources of
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AVAILABILITY

Providing resilience if
one node fails

RICH STATEFUL CLIENTS

The Problem?

The Problem?

IT IS STILL

Very Hard

NO DIFFERENCE

Between a

Slow **NODE**



and a

Dead **NODE**



The network is
INHERENTLY
UNRELIABLE

<http://aphyr.com/posts/288-the-network-is-reliable>

FALLACIES

PETER DEUTSCH'S
8 FALLACIES
OF
DISTRIBUTED
COMPUTING

FALLACIES

PETER DEUTSCH'S 8 FALLACIES OF DISTRIBUTED COMPUTING

1. The network is reliable
2. Latency is zero
3. Bandwidth is infinite
4. The network is secure
5. Topology doesn't change
6. There is one administrator
7. Transport cost is zero
8. The network is homogeneous

So, yes...

So, yes...

IT IS STILL

Very Hard

GRAVEYARD OF DISTRIBUTED SYSTEMS

1. Guaranteed Delivery
2. Synchronous RPC
3. Distributed Objects
4. Distributed Shared Mutable State
5. Serializable Distributed Transactions





DIVIDE & CONQUER

PARTITION
for scale

REPLICATE
for resilience

IMMUTABILITY

“Immutability Changes Everything” - PAT HELLAND

IMMUTABLE Data
SHARE NOTHING Architecture

IMMUTABILITY

“Immutability Changes Everything” - PAT HELLAND

IMMUTABLE Data
SHARE NOTHING Architecture

Is the path towards
TRUE SCALABILITY

THINK IN FACTS

THINK IN FACTS

NEVER DELETE data

Knowledge only GROWS

APPEND-ONLY Event Log

Use Event Sourcing and/or CQRS

THINK IN FACTS

NE

Kn

AP

Use Event Sourcing and/or CQRS



AGGREGATE ROOTS

Can wrap multiple Entities

Aggregate Root is the TRANSACTIONAL BOUNDARY

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STRONG Consistency WITHIN Aggregate

EVENTUAL Consistency BETWEEN Aggregates

AGGREGATE ROOTS

Can wrap multiple Entities

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EVENTUAL Consistency BETWEEN Aggregates

NO LIMIT TO SCALABILITY

REACTIVE APPLICATIONS

GIVES YOU

LOOSE COUPLING

LOCAL STATE

ASYNCHRONOUS
COMMUNICATION

LOCATION
TRANSPARENCY

REACTIVE APPLICATIONS

GIVES YOU

LOOSE COUPLING

LOCAL STATE

ASYNCHRONOUS
COMMUNICATION

LOCATION
TRANSPARENCY

THE TOOLS TO WRITE TRULY
SCALABLE APPLICATIONS

RESILIENCE

**“THE ABILITY OF A SUBSTANCE OR OBJECT TO
SPRING BACK INTO SHAPE.”**

“THE CAPACITY TO RECOVER QUICKLY FROM DIFFICULTIES.”

- MERRIAM WEBSTER

ISOLATE FAILURE

**ERROR
HANDLING**

WORKER

**BUSINESS
LOGIC**

**SEPARATE YOUR BUSINESS LOGIC
FROM YOUR ERROR HANDLING**

FAILURE DETECTION

FAILURE DETECTION

MONITORING USING PROVEN TECHNIQUES

FAILURE DETECTION

MONITORING USING PROVEN TECHNIQUES

NOTIFICATIONS THROUGH DEDICATED CHANNELS

FAILURE DETECTION

MONITORING USING PROVEN TECHNIQUES

NOTIFICATIONS THROUGH DEDICATED CHANNELS

HANDLE PROPAGATION

FAILURE DETECTION

MONITORING USING PROVEN TECHNIQUES

NOTIFICATIONS THROUGH DEDICATED CHANNELS

HANDLE PROPAGATION

HANDLE SCALABILITY

FAILURE DETECTION

MONITORING USING PROVEN TECHNIQUES

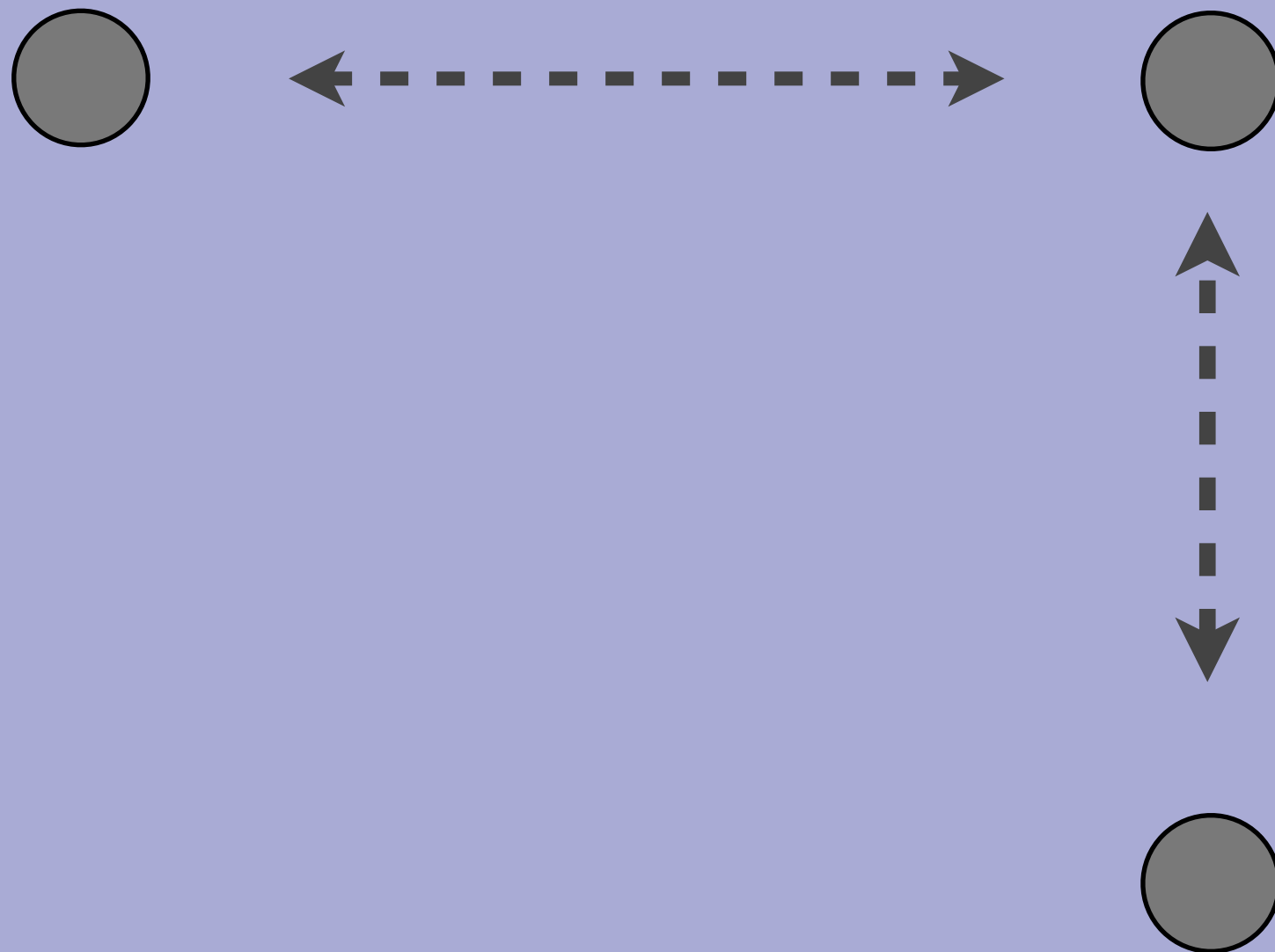
NOTIFICATIONS THROUGH DEDICATED CHANNELS

HANDLE PROPAGATION

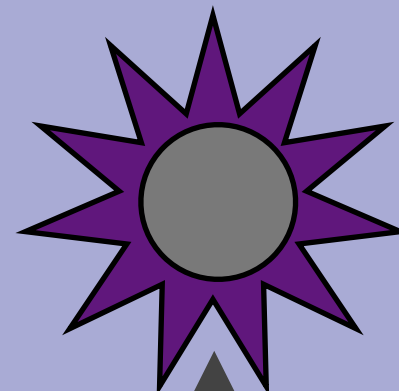
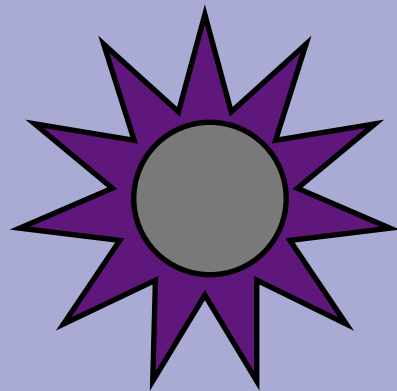
HANDLE SCALABILITY

HANDLE UNCERTAINTY

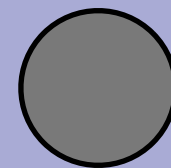
PROPAGATING ERRORS



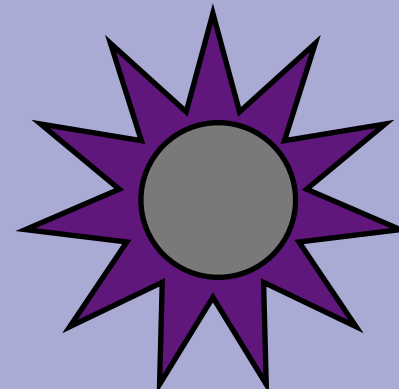
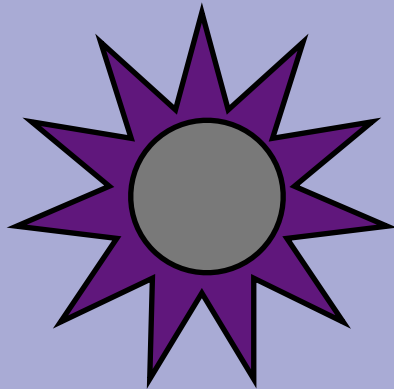
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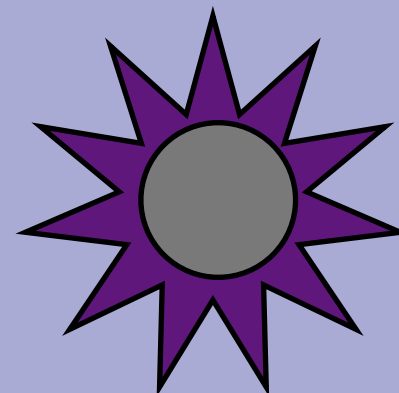
ERROR SIGNAL



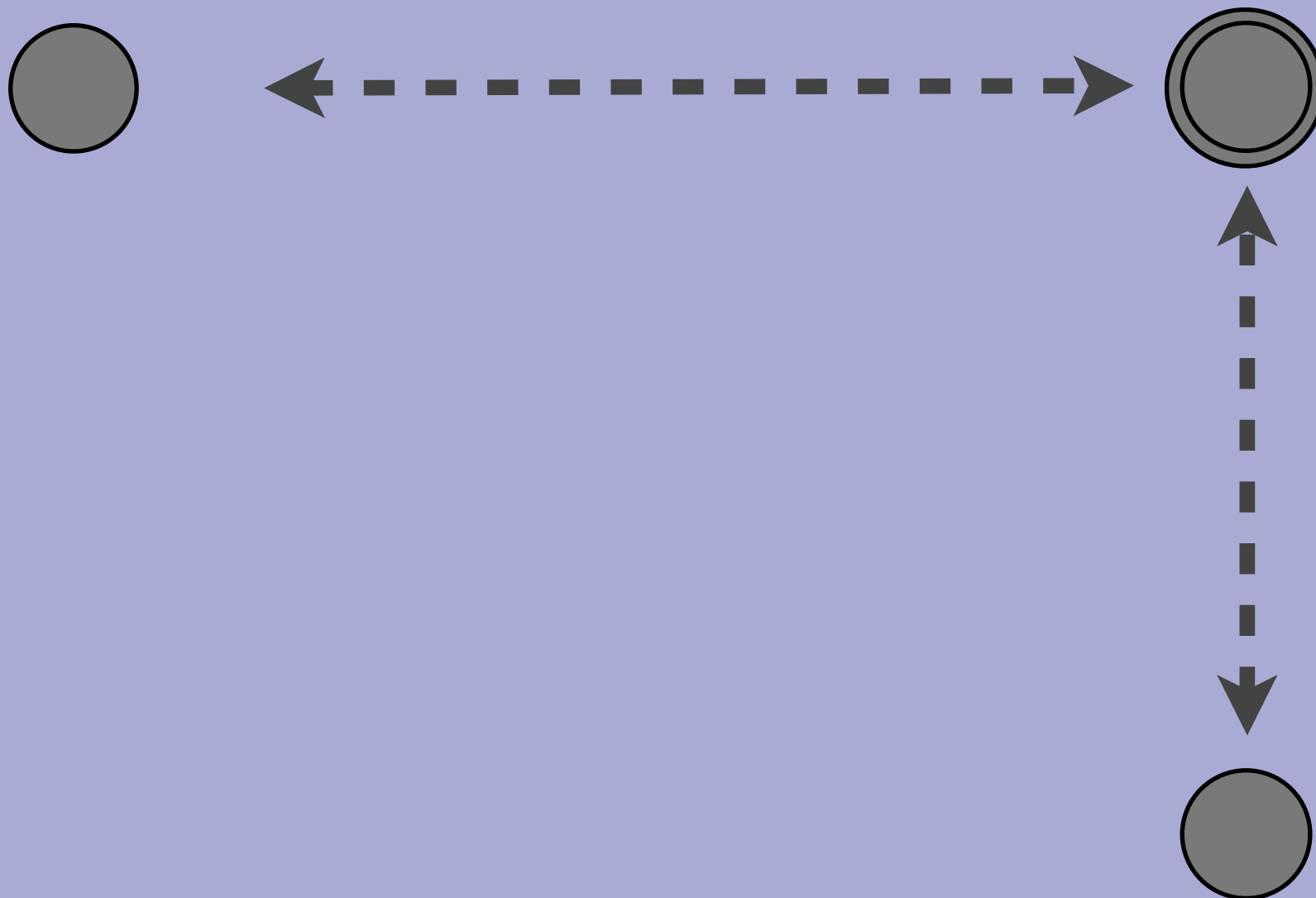
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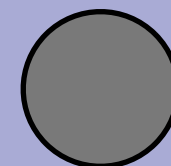
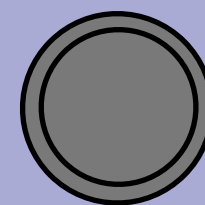
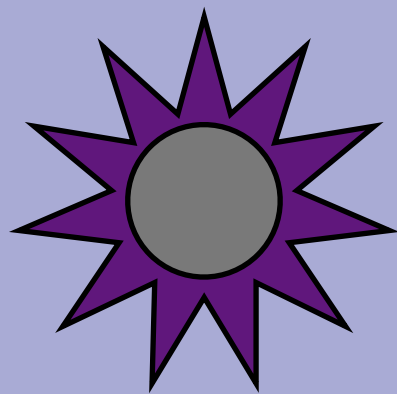
ERROR SIGNAL

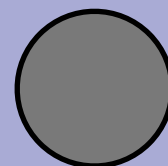
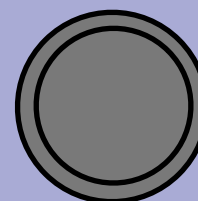
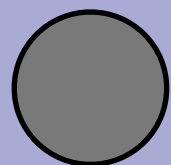


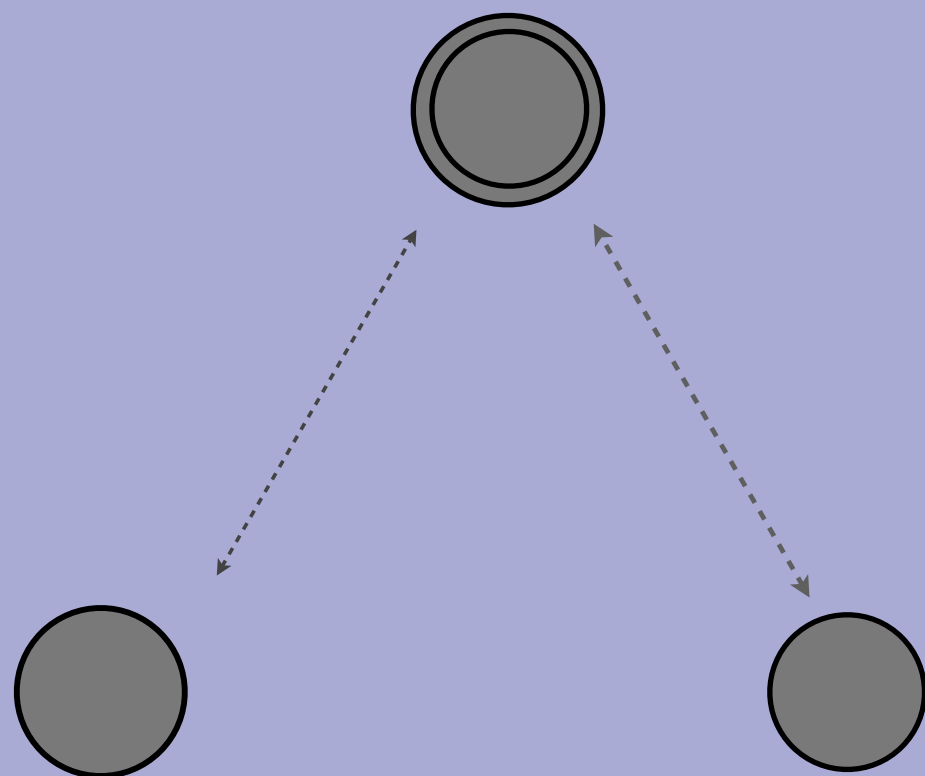
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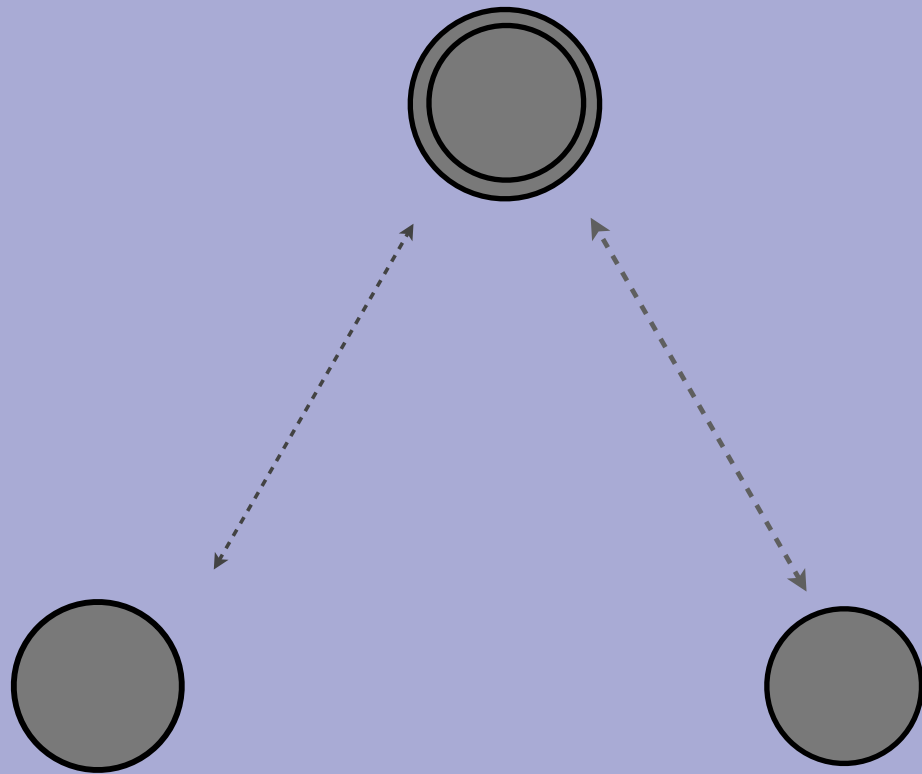
PROPAGATING ERRORS



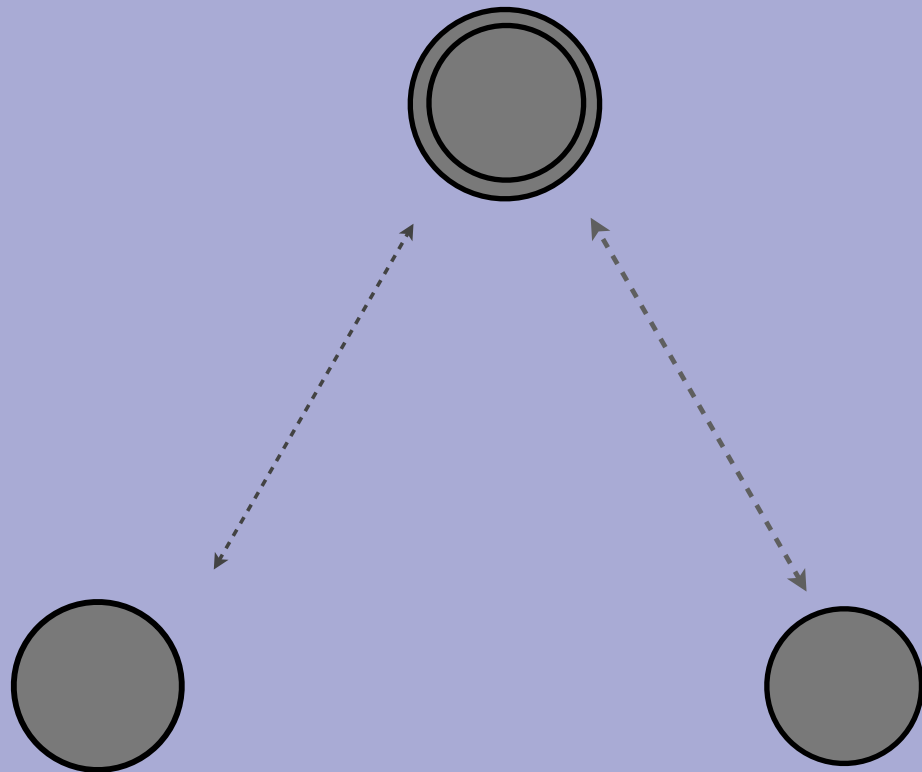




SUPERVISOR

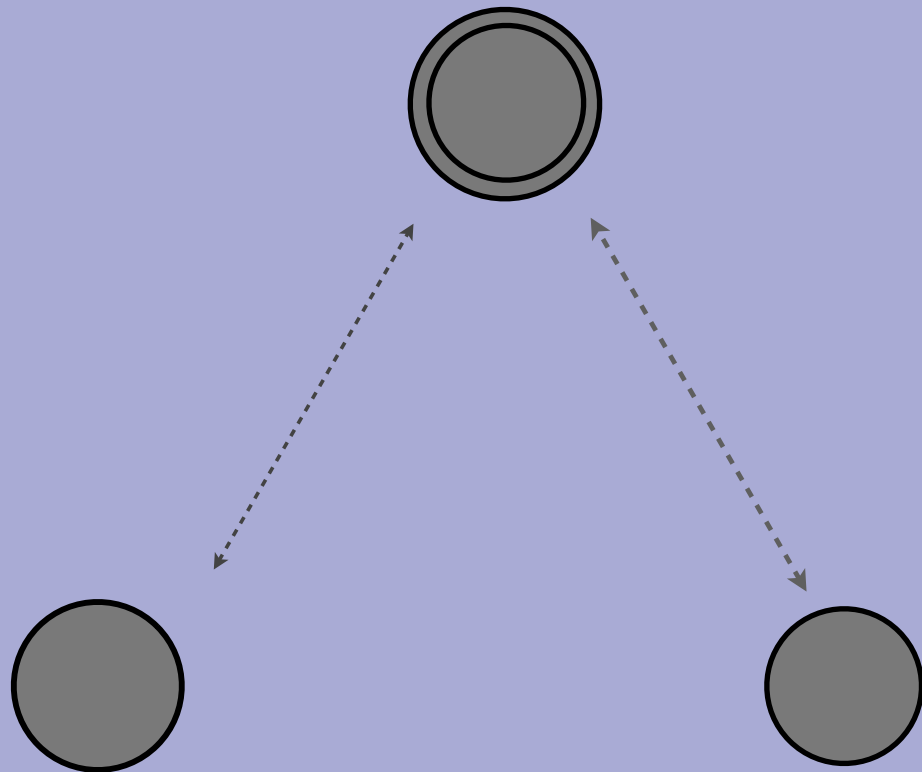


SUPERVISOR



WORKERS

SUPERVISOR



WORKERS

COMPONENT

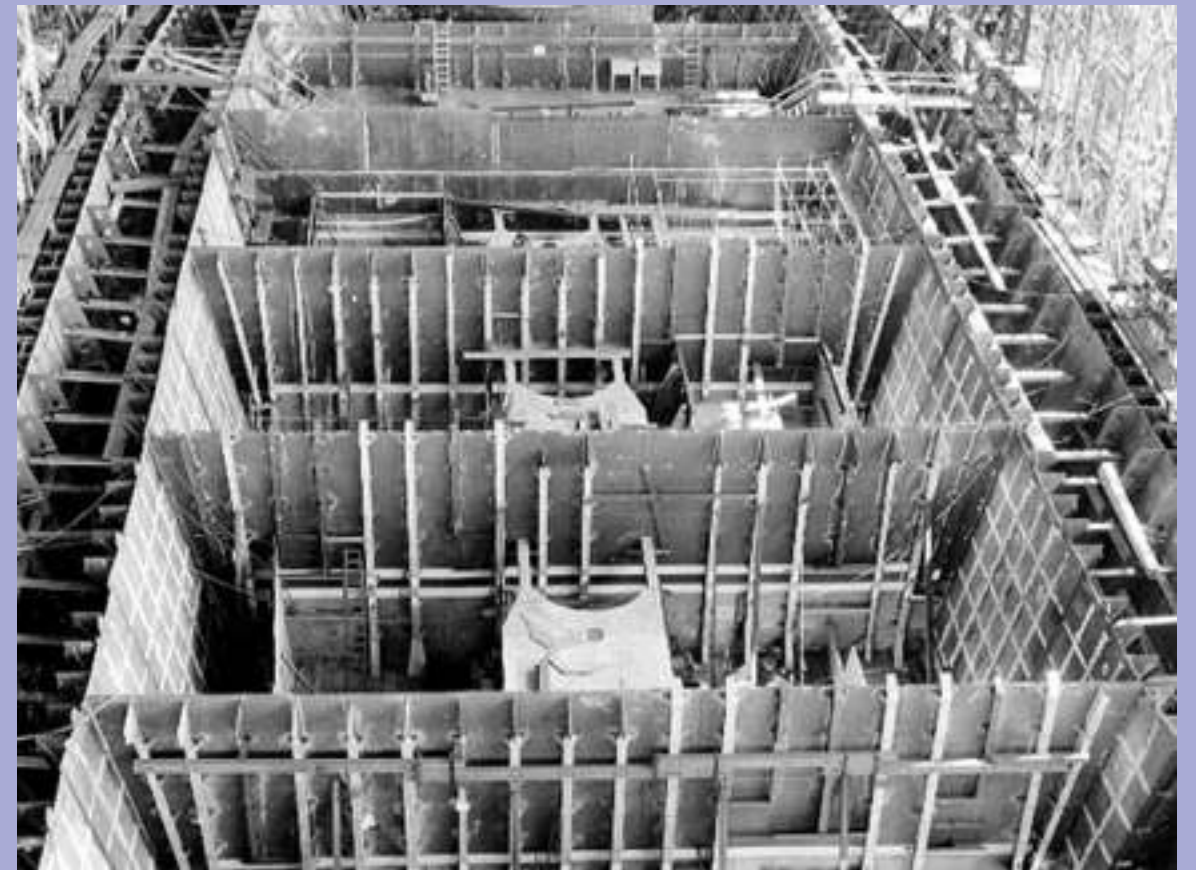
ISOLATING FAILURE

COMPARTMENTALISE

MANAGE FAILURE LOCALLY

ESCALATE ERRORS

AVOID CASCADING FAILURES



RESPONSIVE

“QUICK TO RESPOND OR REACT APPROPRIATELY”

- MERRIAM WEBSTER

KEEP LATENCY PREDICTABLE

UNDER BOTH

- **BLUE SKIES**
- **TRAFFIC BURSTS**
- **FAILURES**

KEEP LATENCY PREDICTABLE

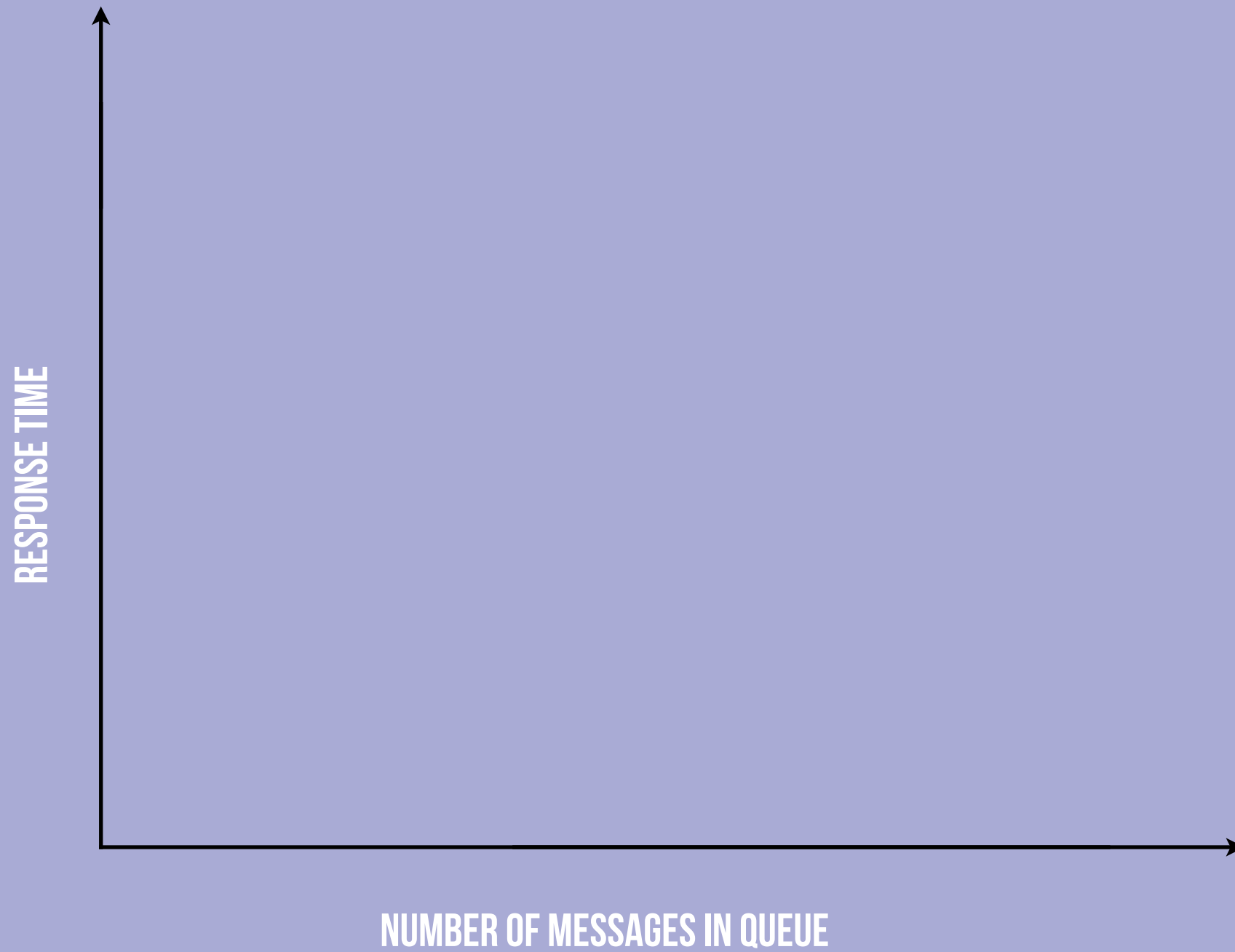
UNDER BOTH

- **BLUE SKIES**
- **TRAFFIC BURSTS**
- **FAILURES**

THE SYSTEM SHOULD

ALWAYS BE RESPONSIVE

USE BACK PRESSURE



USE BACK PRESSURE

RESPECT LITTLE'S LAW:

$$L = \lambda W$$

Queue Length = Arrival Rate * Response Time

RESPONSE TIME

NUMBER OF MESSAGES IN QUEUE

USE BACK PRESSURE

RESPECT LITTLE'S LAW:

$$L = \lambda W$$

Queue Length = Arrival Rate * Response Time

Response Time = Queue Length / Arrival Rate

RESPONSE TIME

NUMBER OF MESSAGES IN QUEUE

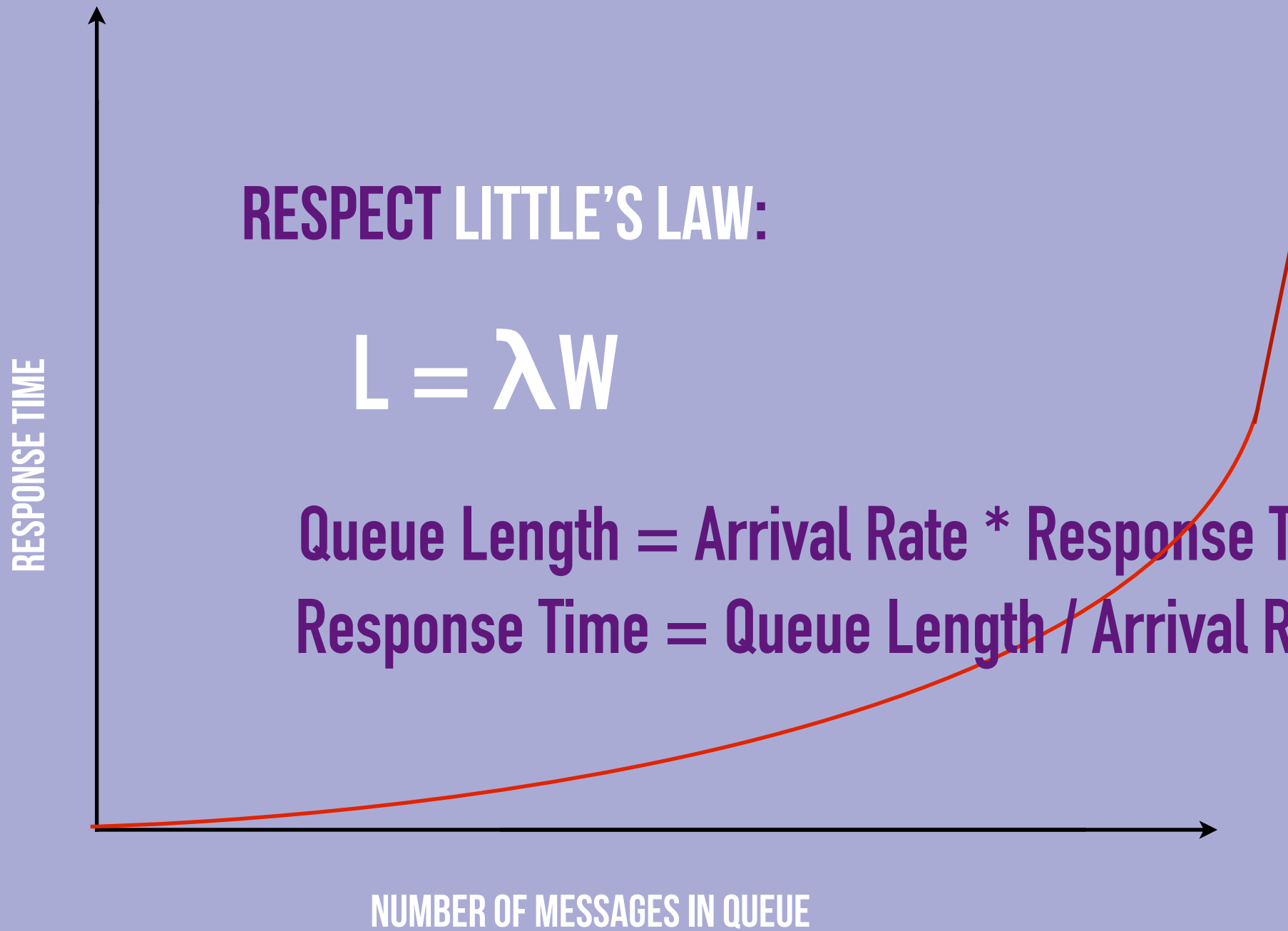
USE BACK PRESSURE

RESPECT LITTLE'S LAW:

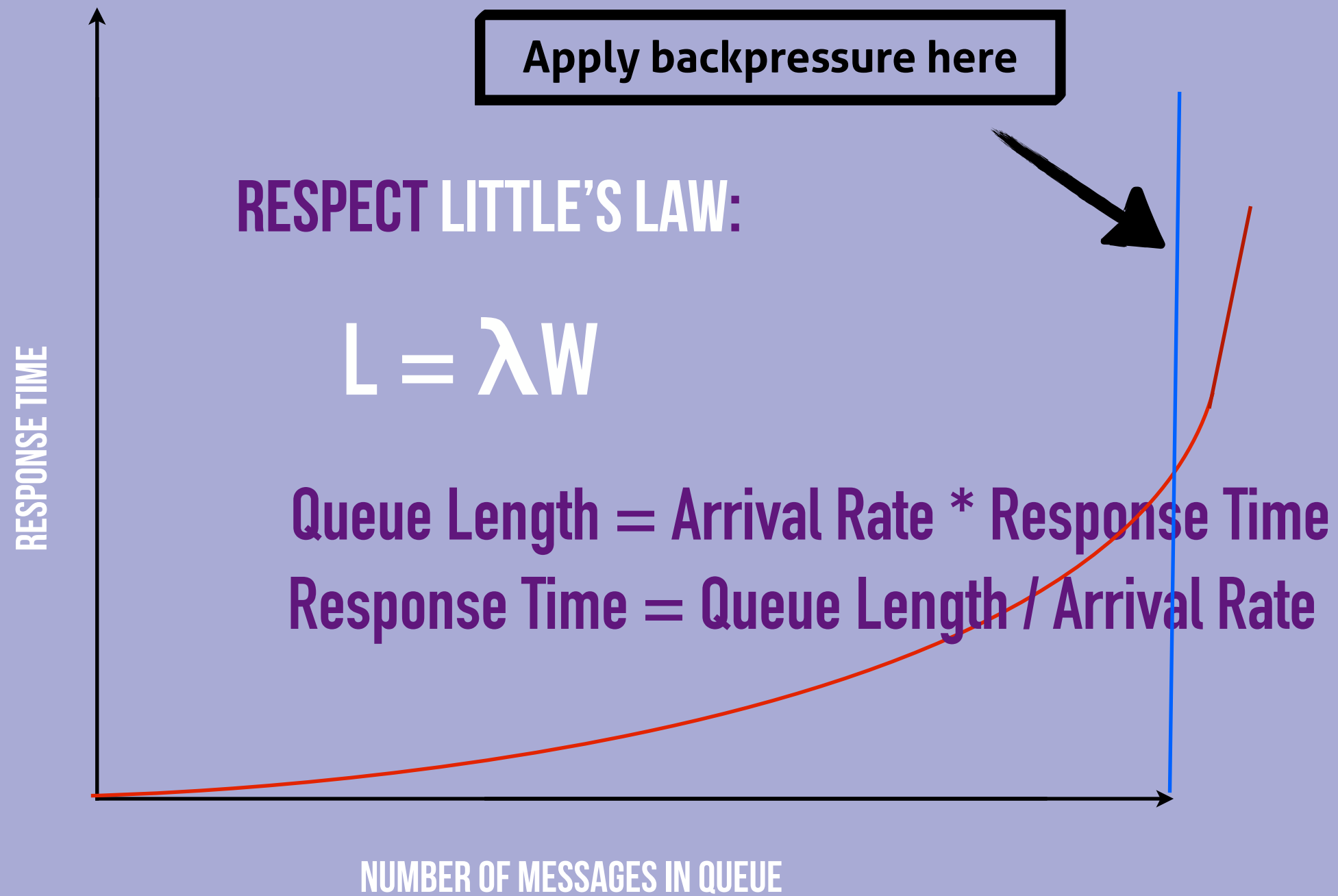
$$L = \lambda W$$

Queue Length = Arrival Rate * Response Time

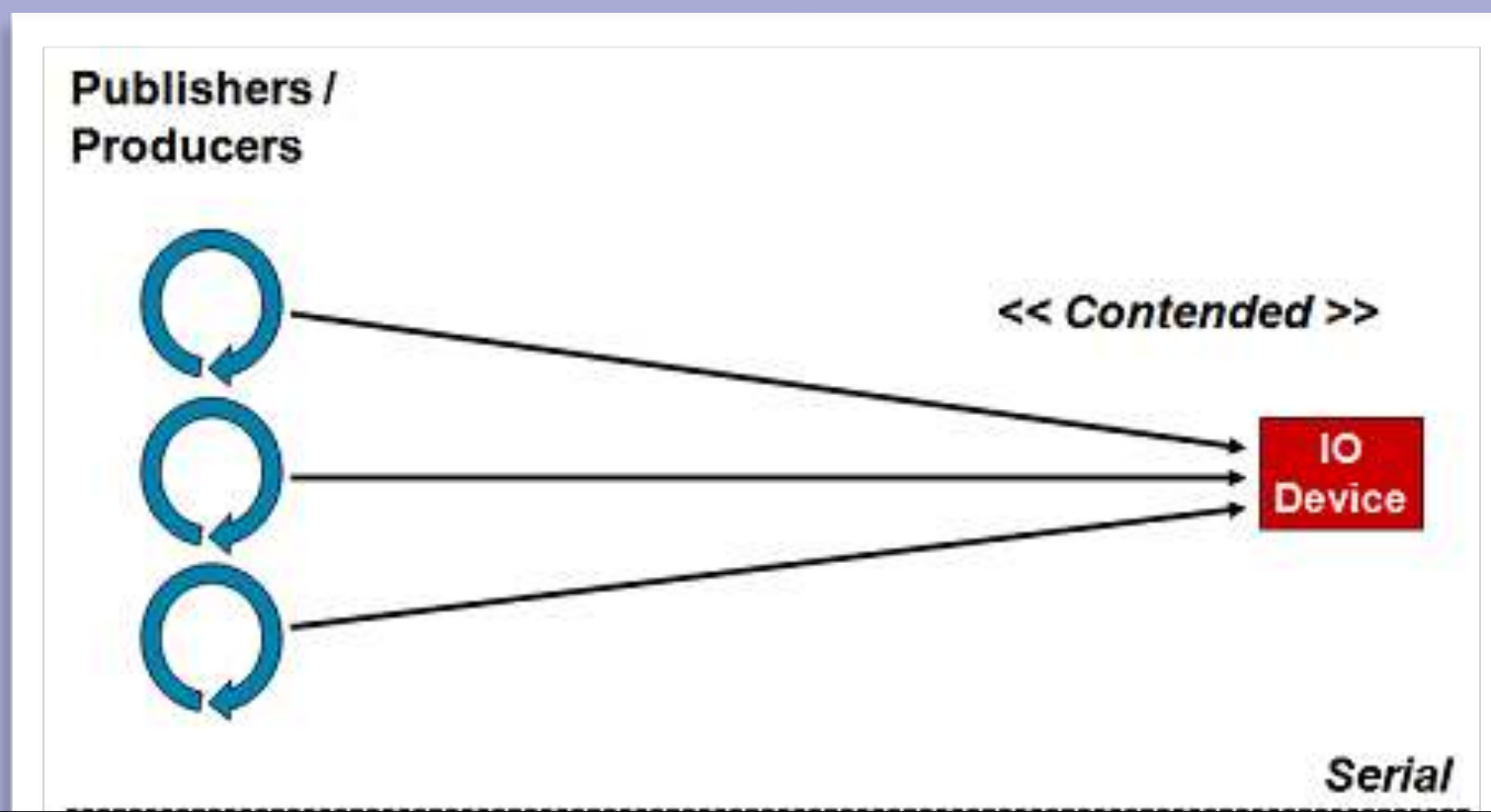
Response Time = Queue Length / Arrival Rate



USE BACK PRESSURE

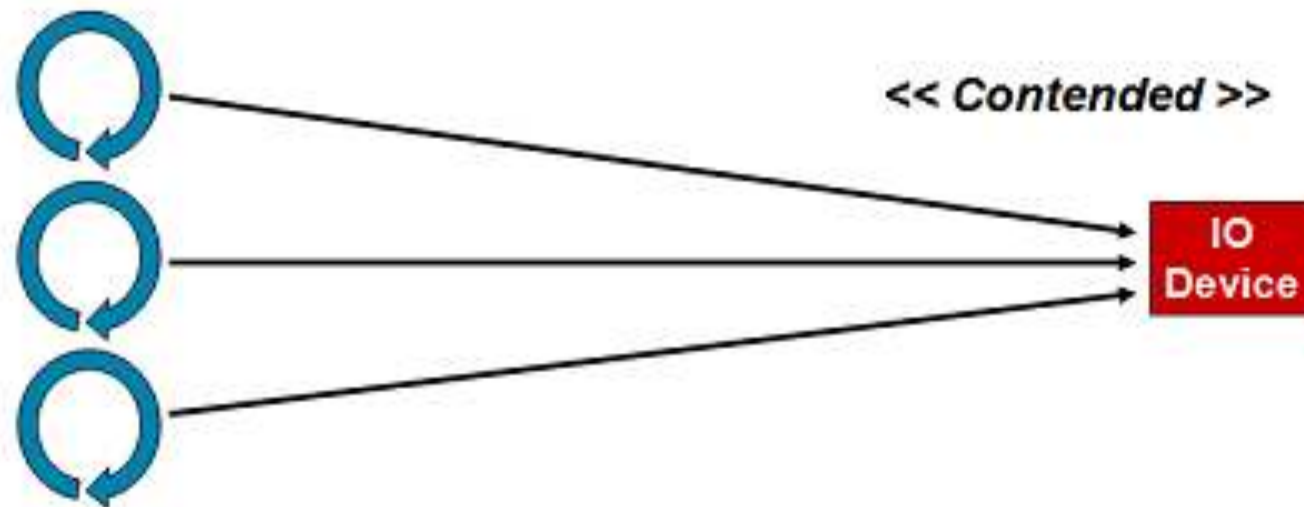


SMART BATCHING



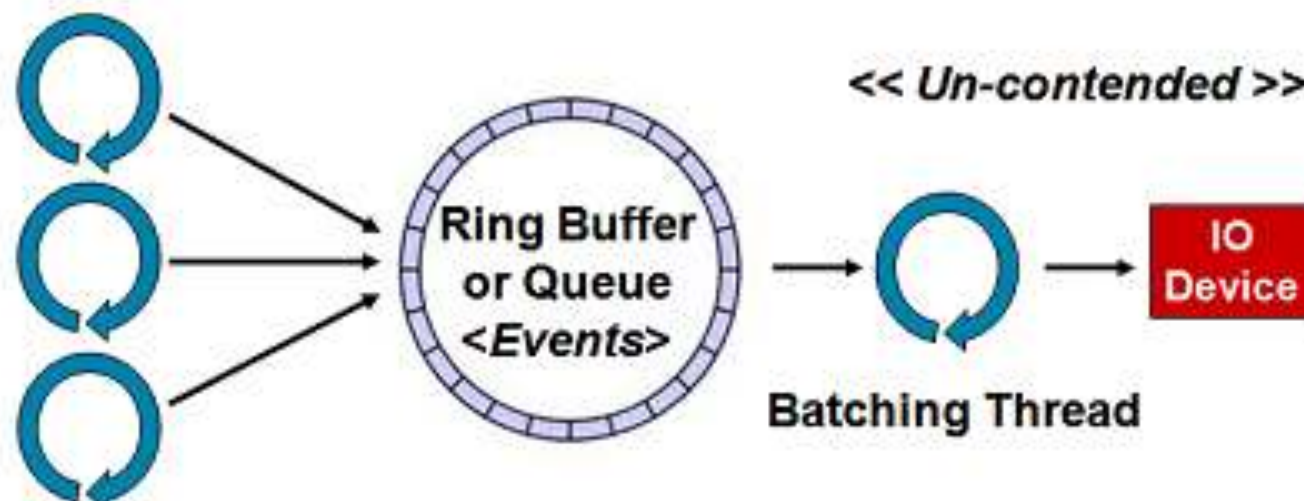
SMART BATCHING

**Publishers /
Producers**



Serial

Batched



REACTIVE WEB & MOBILE APPS

REACTIVE WEB & MOBILE APPS

WE NEED REACTIVE USER INTERFACES

- Interactive
- Data Synchronization
- “Real-time” Collaboration

REACTIVE WEB & MOBILE APPS

1. REACTIVE REQUEST

Async & Non-blocking Request & Response

WE NEED REACTIVE USER INTERFACES

- Interactive
- Data Synchronization
- “Real-time” Collaboration

REACTIVE WEB & MOBILE APPS

1. REACTIVE REQUEST

Async & Non-blocking Request & Response

2. REACTIVE COMPOSITION

Reactive Request + Reactive Request + ...

WE NEED REACTIVE USER INTERFACES

- Interactive
- Data Synchronization
- “Real-time” Collaboration

REACTIVE WEB & MOBILE APPS

1. REACTIVE REQUEST

Async & Non-blocking Request & Response

2. REACTIVE COMPOSITION

Reactive Request + Reactive Request + ...

3. REACTIVE PUSH

Stream Producer

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REACTIVE WEB & MOBILE APPS

1. REACTIVE REQUEST

Async & Non-blocking Request & Response

2. REACTIVE COMPOSITION

Reactive Request + Reactive Request + ...

3. REACTIVE PUSH

Stream Producer

4. 2-WAY REACTIVE

(Bi-Directional Reactive Push)

WE NEED REACTIVE USER INTERFACES

- Interactive
- Data Synchronization
- “Real-time” Collaboration

THE FOUR TRAITS OF **REACTIVE**

RESPONSIVE

SCALABLE

RESILIENT

EVENT-DRIVEN

<http://reactivemanifesto.org>

The Reactive Manifesto

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1. New Challenges for a New Era

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2. Go Reactive

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3. Event-Driven

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5. Resilient

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4. Scalable
5. Resilient
6. Responsive

GO REACTIVE